



CHAIN RULE

Deep Learning

Chain Rule

- The chain rule is a fundamental concept in deep learning that helps calculate the **gradient** of a **neural network's output relative to its parameters**.
- This is important for updating the weights and biases during training.
- The chain rule states that if you have a composite function **$f(g(x))$** , then the derivative of the function with respect to x is the derivative of f with respect to $g(x)$, multiplied by the derivative of g with respect to x .

$$\frac{d}{dx} f(g(x)) = \frac{df}{dg} \cdot \frac{dg}{dx}$$

The Chain Rule Formula

- The chain rule states that if you have a composite function $f(g(x))$, the derivative of this function with respect to x is the derivative of f with respect to $g(x)$, multiplied by the derivative of g with respect to x .
- Mathematically, it is expressed as:
- If $h(x) = f(g(x))$, then

$$h'(x) = f'(g(x)) * g'(x).$$

Here,

- $h'(x)$ denotes the derivative of h with respect to x
- $f'(g(x))$ is the derivative of f evaluated at $g(x)$
- $g'(x)$ is the derivative of g with respect to x .

Applying the Chain Rule

◦ Apply the chain rule, follow these steps:

1. Identify the outer function $f(u)$ and the inner function $u = g(x)$ in your composite function.
2. Compute the derivative of the outer function with respect to its argument, $f'(u)$.
3. Compute the derivative of the inner function with respect to x , $g'(x)$.
4. Multiply the derivatives from steps 2 and 3 to get the derivative of the composite function with respect to x .

Significance of Chain Rule

Gradient computation

- The chain rule allows for the computation of gradients, or derivatives, of a loss function with respect to a network's parameters.

Backpropagation

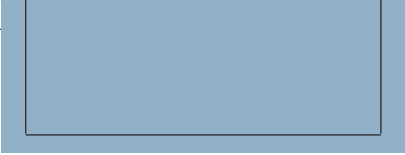
- The chain rule is applied iteratively through the layers of a neural network to propagate gradients backward during backpropagation. This allows the network to learn and adjust its parameters.

Understanding network performance

- The chain rule helps us understand how changes to individual parameters affect the overall performance of the network. This understanding helps the network adjust and improve its predictions.

Key to optimization algorithms

- The chain rule is key to understanding and implementing the optimization algorithms that enable machine learning models to learn from data.



THANK YOU!!